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## Discriminating Voice Pathologies Through a Combination of Spectral and Acoustic Features

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### Abstract

This study compares results from acoustic parameters, spectral parameters, or a combination of both, in classifying speech signals into three classes: healthy speakers (HE), speakers with physiological laryngeal pathologies (PhLP), and speakers with unilateral vocal fold paralysis (UvfP). A Support Vector Machine (SVM) was used for binary and multiclass classification. The highest accuracy rates attained in binary classifications were 94.78% for HE vs. PhLP, 93.35% for HE vs. UvfP, and 73.39% for PhLP vs. UvfP. The highest accuracy rate obtained in the three-class classification was 76.30%, and the highest derived accuracy rates for One vs. All classifications were 92.52% for HE vs. All, 79.79% for PhLP vs. All, and 80.28% for UvfP vs. All. In all cases, the highest accuracy rate resulted from combining acoustic and spectral parameters, demonstrating the advantage of integrating both sets of parameters in healthy and pathological voices discrimination.

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**Keywords:** Pathological voices discrimination; Spectral features; Acoustic parameters; Speech signals.

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## 1. Introduction

Speech is the basis of human communication, constituting one of the main aspects that differentiate us from other living beings. Speech allows us to share our ideas, express our thoughts and feelings, and interact with the world around us in a unique and exclusive way [1]. The tool for this means of communication is the voice, and its quality can impact social life, with estimates suggesting that voice problems affect about 6-15% of the general population. Voice quality can also impact professional life, with estimated prevalence of voice disorders rising to 20-50% among communication professionals such as teachers, potentially affecting up to 80% of this professional class. In more severe cases, voice disorders may indicate health problems with potentially more serious consequences [2] [3].

Various pathologies can impact voice quality, and this study focuses on three of these conditions. The first condition is vocal fold nodules, which are bilateral nodular masses, and the second condition is edema, which is characterized by swellings filled with variable viscosity liquid. Both pathologies are benign mass lesions impacting the same vocal fold structure (Reinke's space), and both pathologies affect the normal movement and vibration of the vocal folds during speech or singing. These two conditions are part of the same set of pathologies known as Physiological Laryngeal Pathologies [4] [5]. The third condition, while not inherently a pathology, often manifests as a symptom of an underlying issue. It is Unilateral Vocal Fold Paralysis and results from damage to the recurrent laryngeal nerve, causing paralysis or immobility of one vocal fold. This condition leads to voice production difficulties such as hoarseness, breathiness, and diminished vocal projection. Consequently, individuals suffering from this condition struggle with articulating speech and may also experience swallowing difficulties [6] [7].

The development of a reliable, automatic method based on speech signal analysis techniques could bring advantages, allowing for a quick, non-invasive diagnosis accessible to everyone at any time. This could serve as a screening tool, to be later complemented by a diagnosis conducted in a traditional manner by specialized personnel [8] [9]. In this context, there are two commonly used approaches, acoustic and spectral analysis, employed in speech signals for healthy and pathological voices discrimination, which will be explored in this study. One approach, acoustic analysis, involves analyzing the evolution of speech signals over time and extracting characteristics using speech processing techniques. The other approach, spectral analysis, is based on the examination of the energy distribution in speech signals across different frequencies. Both approaches consist of two distinct steps: a feature or parameter extraction, and an analysis on these features for healthy and pathological voices discrimination.

Four acoustic parameters, namely fundamental frequency (F0), jitter, shimmer, and Harmonic to Noise Ratio (HNR), were studied in [10]. Speech signals containing the sustained vowels /a/, /i/, and /u/ from 80 healthy speakers (40 of each gender) and 40 pathological ones were used. Parameters were extracted using the Praat software [11], a widely used tool in phonetics and speech science for speech signal analysis. and the mean values were analyzed for the: healthy males, healthy females, and pathological groups. F0 was discarded, as it showed differences between the healthy groups, but it was observed that the three remaining parameters exhibit distinct mean values in healthy and pathological voices, suggesting their potential utility as core parameters for an automated diagnostic system.

In [12], speech signals containing continuous speech or the sustained vowel /a/, from healthy speakers, speakers with physiological laryngeal pathologies, and speakers with Unilateral Vocal fold Paralysis are analyzed. Spectral parameters are extracted from these signals, and applied to different automatic classifiers, namely Support Vector Machines (SVM), Gaussian Mixture Models (GMM), and Fisher Discriminant Analysis (LDA). Regarding the analyses performed on the signals containing the sustained vowel /a/, the highest accuracy rate for classification among the three classes, obtained from different combinations of parameters and classifiers, was 68.2%.

An automatic system for detecting signs of dysarthria in speech signals was proposed in [13], using four different types of parameters: prosodic parameters, jitter, shimmer and HNR, referred to in this study as quality parameters, spectral, and cepstral parameters. Four automatic classification models were tested: SVM, Random Forest, K-Nearest Neighbors (KNN), and Naive Bayes. The best performance was attained using a combination of prosodic and spectral parameters with the Random Forest classifier, achieving an accuracy rate of 94.33%. The SVM classifier achieved the second-best accuracy rate, of 87.13%, when quality parameters and Mel-Frequency Cepstral Coefficients (MFCC) were used as input, suggesting this may be a viable solution for voice disorder detection.

In [14], speech signals from a database are decomposed into ten spectral components using filtering with a bank of ten filters defined on the Mel scale. From these components, averages are calculated, which are then grouped into three clusters, and the relationships between these clusters give rise to three indirect parameters. Six different

automatic classifiers are applied to these parameters for healthy and pathological voices discrimination, achieving average accuracy rates, in the configurations of parameters and classifiers used, of 73% and 75%, respectively.

This study integrates spectral parameters, identical to those used in [16], and acoustic parameters Jitter, Shimmer and HNR, also used in [10]. Spectral parameter dimensionality is reduced using Principal Component Analysis (PCA) and the resultant features are combined with the acoustic parameters. Similar to [12] and [15], a SVM is used with the combined features for healthy and pathological voices discrimination, as well as for pathology discrimination. This study uses an identical dataset as the one used in [12] and [15], focusing exclusively on speech signals containing the sustained vowel /a/, and their results will serve as baseline for the results of this study.

The main objective of this study is to investigate the impact of both spectral and acoustic parameters on the performance of voice pathology discrimination systems of healthy speakers (HE), speakers diagnosed with Physiological Laryngeal Pathologies (PhLP) and speakers diagnosed with Unilateral Vocal Fold Paralysis (UvFP).

This study is structured as follows: the next section presents related work, Section 3 describes the database used in this study, including information about the speech signals, Section 4 outlines the proposed model, and Section 5 presents and discusses the attained results. Finally, the article concludes with Section 6, which summarizes the conclusions drawn from the study.

## 2. Related Work

The dataset used in [12], containing continuous speech or the sustained vowel /a/ speech signals, from healthy speakers, speakers with physiological laryngeal pathologies, and speakers with Unilateral Vocal fold Paralysis are used in another study [15]. Three different features, Mel-Frequency Cepstral Coefficients (MFCC), Line and Mel-Line Spectral Frequencies (LSF and MLSF) are extracted, and these are applied to different automatic classifiers, namely Support Vector Machines (SVM), Gaussian Mixture Models (GMM), and Fisher Discriminant Analysis (LDA). All tests were performed separately for continuous speech and sustained vowel /a/ speech signals.

In the context of discriminating between healthy and pathological voices, the results obtained with continuous speech consistently surpassed those obtained with sustained vowel /a/ signals, indicating that continuous speech is better suited for this task. However, in the discrimination of voice pathologies, the results were less definitive. The authors attributed this ambiguity to the challenge of accurately extracting parameter values from voiced portions of the signal. Despite continuous speech being considered more suitable for this discrimination, the detection of voiced portions might be difficult due to the potential degradation of pathological voices. Regarding the signals containing the sustained vowel /a/, the highest accuracy rate for classification among the three classes, obtained from different combinations of parameters and classifiers, was 70.1% when using LSF parameter and LDA classifier.

In [16], 61 speech signals are used, one per speaker, sampled at a frequency of 22.05 kHz with a minimum duration of two seconds, containing the sustained vowel /a/. The signals are divided into four classes: healthy speakers, speakers with neuromuscular laryngeal pathologies, speakers with Reinke's Edema, and speakers with vocal nodules. The last two classes are combined into a single class named physiological laryngeal pathologies class.

It was discovered that the dimensions related to the frequency ranges of the first formant of the vowel /a/, and of the fundamental frequency and the first harmonic, are the most relevant in this context. In fact, the energy standard deviation dimension within the [147;231] Hz band alone was enough to distinguish between healthy voices and those with physiological pathologies with 100% accuracy, between healthy voices and those with neuromuscular pathologies with 96.55% accuracy, and between the two pathologies with 93.48% accuracy in this dataset. These results validate these spectral parameters as potential inputs for all three discriminations studied.

## 3. Dataset

In this work, a subset of the Massachusetts Eye and Ear Infirmary (MEEI) database [17] was used, The MEEI database was developed by the MEEI Voice and Speech Laboratory and commercialized by the Kay Elemetrics Corp. All ethical issues, data protection, and pathology labeling are responsibility of these companies.

This study uses an identical dataset to the one used in [12] and [15], consisting in speech signals with 25 kHz sampling frequency and 16 bits per sample PCM encoding, containing the sustained vowel /a/. One speech signal per speaker was used, these being adults of both genders, with no separation made in this study regarding age or gender.

Speakers are categorized into four groups: 36 healthy subjects (HE), 59 subjects diagnosed with unilateral vocal fold paralysis (UvFP), and 56 subjects diagnosed with physiological laryngeal pathologies (PhLP) (vocal cord nodules, and edema). In total the dataset is composed by 151 samples.

#### 4. Proposed Model

This study was implemented in Python (version 3.9.12) and the methods employed for signal acquisition and pre-processing, feature extraction, data transformation, classification, and performance evaluation were either custom-built or pre-built functionalities available in Scikit-Learn [18] or Parselmouth [19] libraries.

The proposed model involves the following steps, described as follows.

##### 4.1. Spectral features extraction

Before the spectral features extraction, speech segments are normalized to have their absolute maximum amplitude set to 1, ensuring consistent loudness across speakers. Additionally, silences exceeding 15 ms at the beginning and end of the signals are removed using a process based in [20], to focus the analysis on the voiced portion. The signals are divided into 15 ms frames and scanned from left to right to identify the beginning of speech, where each frame's power is calculated, and its value compared against two thresholds,  $k_1$  and  $k_2$ . If a frame's power exceeds  $k_1$  and the next frame's power exceeds  $k_2$ , that frame is marked as the start of the speech signal, and prior frames are discarded. The same process is applied from right to left to remove silence at the end of the segment.

Spectral parameter extraction follows the approach described in [16]. Speech segments are decomposed into frames of 30 ms, obtained with a 10 ms step size, and the first and last three frames are discarded to mitigate instabilities at the beginning and end of the speech. Each remaining frame undergoes a 4096-point FFT to derive its spectrum and obtain energy bands through filtering. The filter bank used in this process covers the frequency range between 0 (not included) and 4 kHz and consists of 20 filters with triangular shape and unit area frequency responses. These filters are evenly spaced and have equal bandwidths within the Mel scale [21]. This choice ensures that the resulting filter bank covers the 4 kHz frequency range with consistent resolution in terms of perceived pitch.

The filtering process generates an array of 20 by  $N$  energy values for each speech signal, where 20 corresponds to the number of filters and frequency bands, and  $N$  is the number of frames analyzed in each segment. The value of  $N$  varies depending on the duration of the speech signal. From each array, the means and standard deviations are calculated for each of the 20 frequency bands, resulting in two arrays of 20 values each. One array contains the mean energies, while the other contains their respective standard deviations, for each speech signal.

As noted in [16], higher frequency energy is not useful in distinguishing between healthy and pathological voices, so the last 8 dimensions of both arrays are discarded. Also as noted in [16], the most relevant information for these discriminations lies in the energy of lower frequencies, corresponding to the fundamental frequency and first harmonics, and intermediate frequencies, corresponding to the first formant of the vowel /a/. Consequently, each array is split into two: one focusing on lower frequencies and the other on intermediate frequencies.

An overview of the spectral parameter extraction process is depicted in Figure 1.

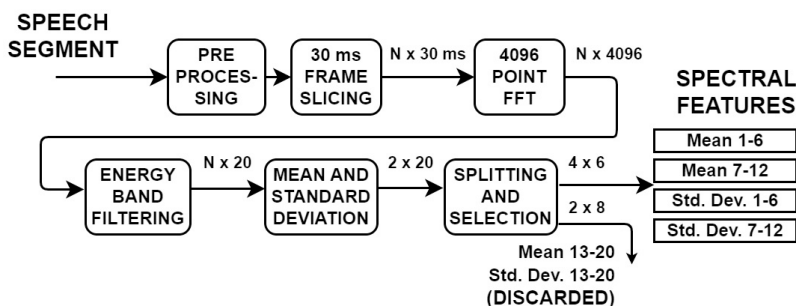


Fig. 1. Spectral features extraction process.

At this point, each speech segment's spectral features are represented by a 24-dimensional variable consisting in four 6-dimensional arrays.

#### *4.2. Acoustic features extraction*

For the acoustic features extraction process, a third-party library called Parselmouth [19] is used. Parselmouth is a Python library built on top of the Praat software, bringing some of Praat's functionality into the Python environment.

Using a third-party library means its operation is not transparent, accessible, or modifiable to the user. When the extraction of acoustic features failed in certain speech signals, it resulted in the exclusion of those signals, reducing the original subset of 154 signals to the 151 utilized in this study, as previously referred in Section 3.

Three acoustic parameters are extracted: Jitter, Shimmer, and Harmonic-to-Noise Ratio (HNR). Jitter quantifies the variation in time intervals between consecutive pitch periods and is expressed as a percentage of the average pitch period. Shimmer quantifies the variation in amplitude between consecutive glottal cycles and is expressed as a percentage of the average glottal pulse amplitude. HNR quantifies the ratio of harmonic components (related to the fundamental frequency of the voice signal) to non-harmonic or noise components in the signal.

Acoustic parameters were combined after the extraction and were always used as a 3-dimensional variable.

#### *4.3. Spectral and acoustic features combination*

Before combining the two sets of parameters, the dimensionality of the spectral parameters is reduced.

As noted in [16], energy at lower frequencies contains more relevant information than that at intermediate frequencies. Therefore, the arrays corresponding to lower frequencies are reduced to 2 dimensions, and those corresponding to intermediate frequencies to 1 dimension. This dimensionality reduction is achieved using Principal Component Analysis (PCA), which is a linear transformation technique that re-maps the data into a new coordinate system in which the new variables, known as principal components, are linear combinations of the original ones. The first principal component (PC) is the linear function which has the maximum possible variance, the second PC is the linear function with maximum possible variance subject to being uncorrelated with the first PC, and so on. So, applying PCA and reducing data dimensionality by retaining the first principal components (discarding the rest), maximizes the information preserved in the process [22].

Finally, the spectral parameters from the four resulting arrays (two with 2 dimensions and two with 1 dimension) are combined with the three-dimensional variable containing the acoustic parameters, resulting in a joint 9-dimensional variable per speech segment.

#### *4.4. Classification*

The classification tasks are performed using Support Vector Machine (SVM) classifiers, also used in related studies [12], [14] and [15]. Three other classifiers, Fisher Discriminant Analysis, Random Forest and K-Nearest Neighbors were tested, but they showed poorer performance in preliminary tests, so their use was discarded. SVM classifier works by finding the optimal hyperplane that best separates different classes in the dataset. This hyperplane is chosen to maximize the margin, which is the distance between the hyperplane and the nearest data points from each class, also known as support vectors [23]. A radial basis function (RBF) kernel was used, as this kernel consistently outperformed other available kernels in all preliminary tests.

To optimize the use of SVM, considering that the results of this classifier are sensitive to dimensions with different scales, the data are transformed to have a mean of 0 and a standard deviation of 1. Just like with dimensionality reduction using PCA, this transformation is applied before training the classifier and the model is fitted with the training data. Then, the transformation is applied separately to both the training and test data. This approach ensures that data pre-processing doesn't bias the results when evaluating the classifier's performance.

The results are obtained through cross-validation, a technique that involves dividing the data into mutually exclusive subsets (folds) of equal size. One of these folds is used for testing, while the remaining ones are used for training the classifier, ensuring that no data is used in the training and test sets simultaneously. This process is repeated until all folds had taken their turn as the test set, ensuring that every data point is included in the testing

phase. In this study, cross-validation is performed with 5 folds, maintaining the same class instances ratio as the original data in all folds, resulting in a test subset of 20% and a corresponding training subset of 80% of the total data. To obtain more reliable results, cross-validation was iterated 1000 times, with shuffle enabled and a unique random state for each iteration to ensure distinct sets of folds for every performed evaluation. The reported values represent the average across all iterations for each classification.

#### 4.5. Metrics and results comparison

The performance of all classification tasks is evaluated using the Accuracy (ACC) metric, which measures the ratio of correct predictions to the total number of instances.

Results from studies [12] and [15], against which the results of this study will be compared, are represented by Accuracy only for classification among three classes. In binary One vs. All classifications, results are represented by the True Positive Rate (TPR), which quantifies the proportion of correct predictions within a class relative to the total instances of that class, and the Positive Predictive Value (PPV), which indicates the proportion of correct predictions within a class relative to all predictions made for that class.

For comparison of results, it is essential to use the same metric. Given that TPR, PPV, total instances (TOTAL) and individual class instances (TOTCLASS) values are known, Accuracy for a given class  $i$  vs. All binary classification can be calculated from these values as shown in Equation (1):

$$ACC_i = \frac{TOTAL - \left[ \left( \frac{1 - TPR}{TPR} \right) \times \left( \frac{1 - PPV}{PPV} \right) \times TOTCLASS_i \times TPR \right]}{TOTAL} \quad (1)$$

This expression will be used to convert TPR and PPV results from [12] and [15] into Accuracy values to be compared with this study's results.

## 5. Results and Discussion

Three One vs. One binary classifications were performed, involving two of the three existing classes in each comparison. Each classification was performed three times, with the first one using only acoustic parameters, the second classification using only spectral features, and the third one using the combination of both feature sets.

A multiclass classification, involving all three classes, was also conducted three times: once using only acoustic parameters, another using only spectral parameters, and the third one using the combination of both parameter sets.

Results from the 3-class classification allowed for the derivation of accuracy rates for the three binary One vs. All classifications, associated with the existing classes. This involved considering one class as Positive and combining the other two into the Negative class. These accuracy rates, along with those obtained from the 3-class and the binary One vs. One classifications, are presented in Table 1. Additionally, the accuracy rates from studies [12] and [15], calculated using Equation 1, are included for comparison. All values are in percentage.

Table 1. Accuracy results for One vs. One, One vs. All and 3-class classifications (results in [%]).

| Classification | HE vs. PhLP  | HE vs. UfvP  | PhLP vs. UfvP | 3-Class      | HE vs. All   | PhLP vs. All | UfvP vs. All |
|----------------|--------------|--------------|---------------|--------------|--------------|--------------|--------------|
| Acoustic       | 85.52 ± 2.57 | 90.69 ± 1.96 | 68.13 ± 2.88  | 66.86 ± 2.00 | 88.25 ± 1.09 | 69.51 ± 2.03 | 75.97 ± 1.77 |
| Spectral       | 86.53 ± 3.48 | 85.85 ± 3.01 | 65.39 ± 5.63  | 66.68 ± 4.07 | 88.43 ± 2.14 | 73.07 ± 4.29 | 71.87 ± 4.23 |
| Combined       | 94.78 ± 2.19 | 93.35 ± 1.60 | 73.39 ± 3.97  | 76.30 ± 3.30 | 92.52 ± 1.88 | 79.79 ± 3.45 | 80.28 ± 3.12 |
| ACC [12]       | n/a          | n/a          | n/a           | 68.2         | 87.0 *       | 73.4 *       | 76.0 *       |
| ACC [15]       | n/a          | n/a          | n/a           | 70.1         | 90.1 *       | 74.7 *       | 79.2 *       |

\* These values were calculated using Equation (1)

The accuracy rates obtained in this study are represented in Table 1 by the mean value with a 95% confidence interval margin. The results in Table 1 pertaining to studies [12] and [15] are exclusively for tests conducted using speech signals containing sustained vowel /a/. For the 3-class classification, the values were directly extracted from these studies. For One vs. All classifications, they were calculated using Equation 1. These studies do not include results for One vs. One classifications.

Table 1 reveals that employing a combination of acoustic and spectral parameters results in higher accuracy rates compared to using either parameter set individually. The accuracy rates show improvements ranging from 2.66% to 10.18% when using the combined parameter sets compared to using acoustic parameters alone, and from 4.09% to 9.44% compared to using spectral parameters alone. This accuracy rate improvement is observed across all classifications, from which it can be inferred that the combination of acoustic and spectral parameters consistently enhances the relevant information for distinguishing the three classes in the speech signal.

The results attained when using acoustic or spectral features alone, suggest that spectral features are the most relevant for PhLP class discrimination, while the acoustic features are more relevant for UvFP class identification. Accuracy rates obtained with acoustic features are higher than those attained with acoustic features for HE vs. UvFP classification (90.69% vs. 85.85%) as well as for UvFP vs. All classification (75.97% vs. 71.87%). This can be explained by the fact that acoustic parameters primarily contain information regarding vocal fold vibration, specifically the glottal pulse, during speech production. Paralysis of a vocal fold, as experienced by speakers in the UvFP class, is expected to have a more significant impact on the glottal pulse compared to a lesion on the same tissue, as observed in speakers from the PhLP class.

On the other hand, accuracy rates obtained with spectral features yield higher values than those attained with acoustic features for HE vs. PhLP classification (86.53% vs. 85.52%) as well as for PhLP vs. All classification (73.07% vs. 69.51%). These results suggest that spectral parameters are more significant for the PhLP class discrimination. This aligns with what was observed in [16], which used similar spectral parameters, where, through inspection of energy bands, it was observed that speakers with physiological laryngeal pathologies exhibited greater vocal instability, reflected in the variation of energies in frequency bands related to pitch and first harmonics.

When comparing the accuracy rates obtained in this study with those from [12] and [15], our study consistently shows higher values, with improvements ranging from 1.08% to 8.1%, proving to be advantageous combining acoustic and spectral features for all performed classifications. However, when comparing the results using only one set of features, accuracy rates obtained in [15] are consistently higher, suggesting that using a different set of spectral features and a different classification model could lead to improved results. Additionally, it's worth noting that both [12] and [15] concluded that continuous speech signals are better suited for extracting spectral features than sustained vowel /a/ speech segments. Therefore, using continuous speech signals for spectral feature extraction may also lead to performance improvements.

## 6. Conclusions

In this study, classifications were conducted on speech signals belonging to three distinct categories: healthy speakers (HE), speakers with physiological laryngeal pathologies (PhLP), and speakers with unilateral vocal fold paralysis (UvFP), based on acoustic parameters, spectral parameters, and a combination of both parameter sets.

A total of seven classifications were conducted, including three binary One vs. One, one 3-class, and three binary One vs. All classifications. Across all classifications, combining acoustic and spectral parameters consistently outperformed using only one parameter set, evidencing the advantage of integrating both types of features and suggesting potential benefits in similar classification procedures.

When using a single set of features, spectral parameters demonstrated the greatest influence in the PhLP class discrimination as these consistently achieved higher accuracy rates compared to using only acoustic ones for this class discrimination. In contrast, for the UvFP class, acoustic parameters were found to be the most influential, consistently yielding higher accuracy values than spectral features. These findings suggest that there may be advantages in implementing ensemble or hierarchical classifiers that use acoustic and spectral parameters separately.

When combined features were used, the results consistently outperformed the sustained vowel /a/ based results of [12] and [15], for all classifications, reinforcing the advantage of combining acoustic and spectral parameters.

However, when comparing results using only one set of parameters, accuracy rates were lower than those obtained in [15], suggesting potential improvements with a different set of spectral features or a different classifier model.

As future work, we propose implementing ensemble and hierarchical classifiers, incorporating additional acoustic parameters and different spectral features into the model, and testing it on different datasets.

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